



Piezoresistive Pressure Transmitter



General Industrial Applications

Features

- Measuring ranges from 100mbar to 600bar
- Absolute, gauge and sealed gauge
- Accuracy: $\pm 0.25\% \text{FSO}$ or $\pm 0.5\% \text{FSO}$
- Calibrated and temperature compensated
- Stainless steel construction
- Piezoresistive pressure sensor design
- Variety of Pressure & Electrical connections
- Output 4...20mA, 0...10V, 0...5V and others

Product Overview

The pressure transmitter is made from high-quality silicon piezoresistive sensor.

The piezoresistive sensor is packaged in stainless steel housing.

It is precision engineered to fit most industrial pressure measurement.

The compact and rugged design makes these pressure transmitter suitable for applications including process control systems, hydraulic systems and valves, refrigeration and HVAC controls, level measurement and test equipment .

Awide range of process connection and electrical connection options are available to meet almost requirement.

Applications

- Process control systems
- Refrigeration and HVAC controls
- Hydraulic systems and valve
- Machine building
- Pumps and compressors

Standard Pressure Ranges

Nominal pressure	gauge	sealed gauge	absolute
-1...0bar	•		
-0.35...0bar	•		
-0.2...0bar	•		
0...0.1bar	•		
0...0.2bar	•		
0...0.35bar	•		•
0...0.7bar	•		•
0...1bar	•		•
0...1.6bar	•		
0...2.5bar	•		•
0...4bar	•		•
0...6bar	•		•
0...10bar	•	•	•
0...16bar	•	•	•
0...25bar	•	•	
0...60bar		•	
0...100bar		•	
0...250bar		•	
0...400bar		•	
0...1000bar		•	

Other pressure ranges available. Please consult the factory.

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Performance Specifications

Parameter	Value	Units	Notes
General			
Pressure Range	-0.01-0 -1000	Bar	1bar=14.5Psi
Overpressure	1.5xFS	bar	

Environmental			
Operating Temperature Range	-20 to +85	°C	-4 °F to 185 °F
Compensated Temperature Range	-10 to +70	°C	14 °F to 158 °F
Storage Temperature Range	-40 to +125	°C	-40 °F to 257 °F
Vibration	10	g	20 to 2000Hz
Shock	100	g	10ms
Cycles	10x10 ⁶	cycles	

Electrical @ 25°C(77°F)					
Output Signal	4...20mA	0...5Vdc	1...5Vdc	0...10Vdc	0.5...4.5Vdc(ratiometric)
Power Supply(Vs)	12....36Vdc	12....36Vdc	12....36Vdc	15....36Vdc	5Vdc
Load Resistance	<(Vs-12)/0.02A (For current output), >10kΩ (For voltage output)				
Insulation Resistance	100MΩ @50Vdc				

Physical Specifications	
Media Compatibility	All media compatible with 316L stainless steel
Housing	304 or 316 stainless steel
Diaphragm	316L stainless steel
Seal Ring	Viton or NBR
Oil Filling	Silicone oil
Protection	IP65(Standard), IP68(only for cable outlet)
Net Weight	Approx.185g

Parameter	Minimum	Typical	Maximum	Units	Notes
Performance					
Accuracy	0.1	0.25	0.5	%FSO	1,2
Temp Coeff - Zero		±0.75	±1.5	%FSO	3
Temp Coeff - Span		±0.75	±1.5	%FSO	3
Long-Term Stability		±0.2	±0.3	%FSO/year	1

Notes

1. All values measured at 25°C(77°F)
2. Including non-linearity, hysteresis and repeatability.
3. -10°C to 70°C(14°F to 158°F) with reference to 25°C(77°F).

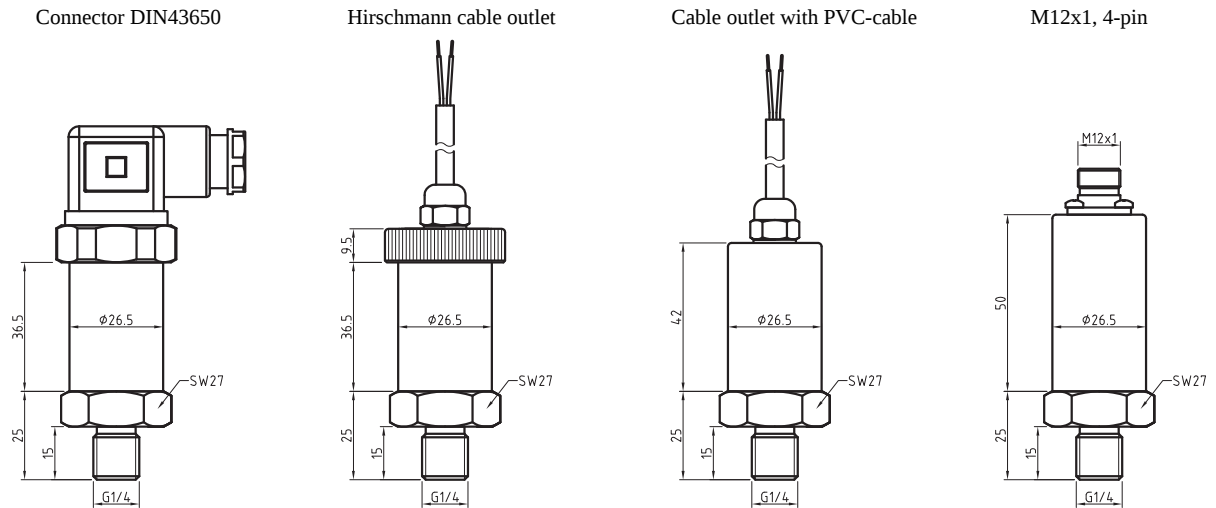
The listed specifications and dimensions are subject to change without prior notice.

Connection Diagrams

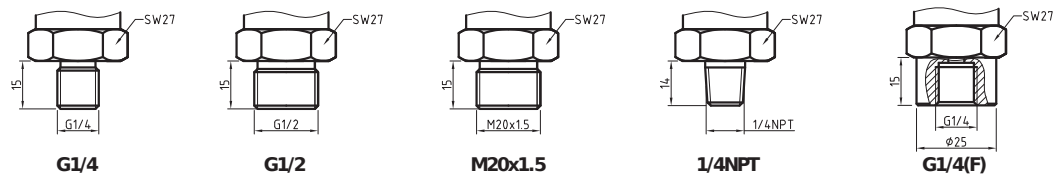
Connector DIN43650			Cable outlet		
	2-wire(current)		2-wire(current)		3-wire(voltage)
	Supply+	1	Supply+	red	red
	Signal+	2	Signal+	yellow	blue
	Gnd	-	Gnd	-	yellow
Connector M12x1(4-pin)					
	2-wire(current)				
	Supply+	1 brown			
	Signal+	3 blue			
	Gnd	-			

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Dimensions (in mm)



Mechanical Connection (in mm)



Ordering Information

Option1: Model						
QYB100	Piezoresistive Pressure Transmitter					
	Option2: Pressure Ranges					
	N001	-1...0bar	0016	0...1.6bar	0600	0...60bar
	N002	-0.35...0bar	0025	0...2.5bar	1000	0...100bar
	N003	-0.2...0bar	0040	0...4bar	2500	0...250bar
	0001	0...0.1bar	0060	0...6bar	4000	0...400bar
	0003	0...0.35bar	0100	0...10bar	6000	0...1000bar
	0007	0...0.7bar	0160	0...16bar	Cxxx	Customized range
	0010	0...1bar	0250	0...25bar		
	Option3: Pressure Type					
	G	gauge				
	A	absolute				
	S	sealed gauge				
	Option4: Output Signal					
		42	4...20mA			
		05	0...5Vdc			
		15	1...5Vdc			
		10	0...10Vdc			
		45	0.5...4.5			
	Option5: Accuracy					
		02	0.25%FSO			
		05	0.5%FSO			
	Option6: Electrical Connection					
		D	Connector DIN43650			
		H	Hirschmann cable outlet,length=1.5m			
		C	Cable outlet with PVC-cable,length=1.5m			
	M	M12x1, 4-pin				
Option7: Mechanical Connection						
	M2	M20x1.5(male)	N1	1/4NPT(male)		
	G4	G1/4(male)	F4	G1/4(female)		
	G2	G1/2(male)	Nx	Customized		